

Thorough Characterization of the Emerging Pathogen *Trichophyton indotineae*: Resistance Profile and MALDI-TOF MS Identification

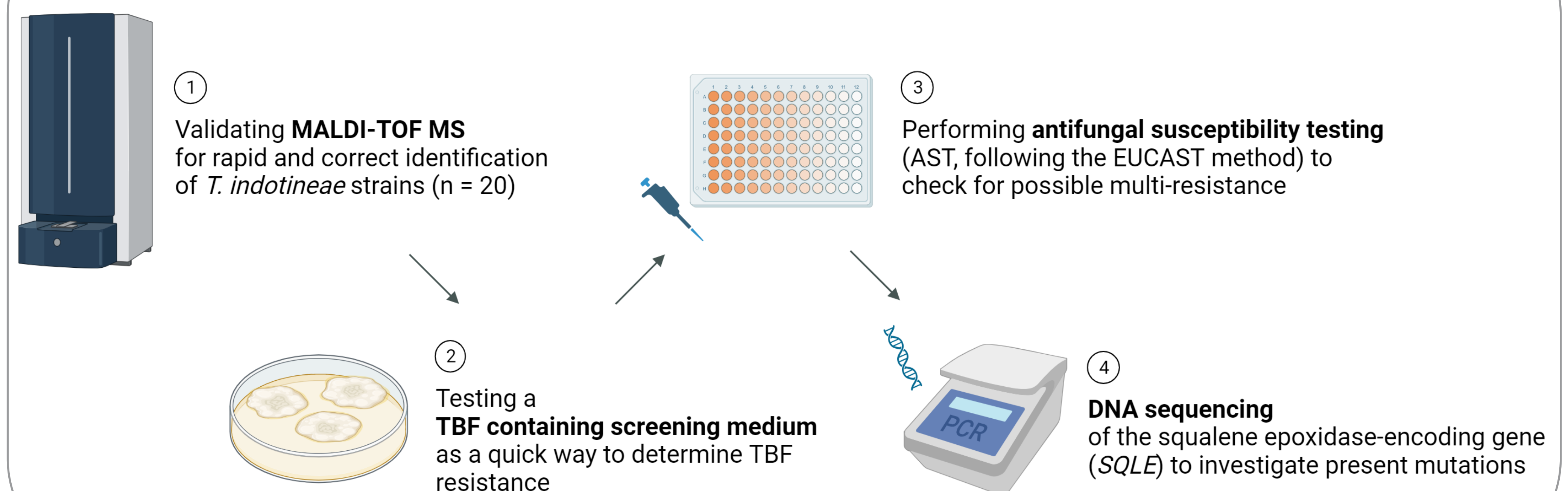
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INTRODUCTION

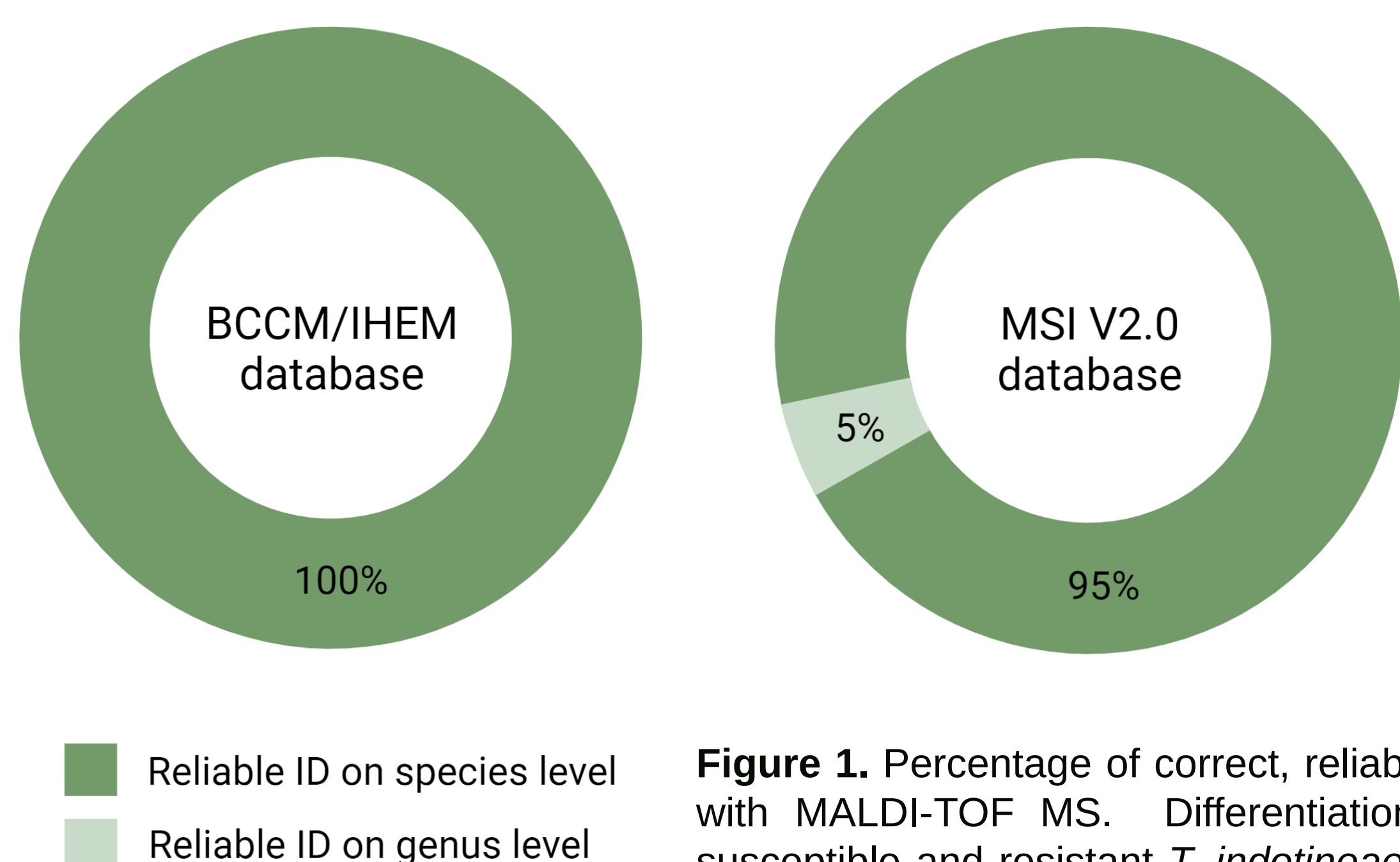
The emerging pathogen, *Trichophyton indotineae* is known to cause severe dermatophytoses such as *tinea corporis* and *tinea cruris*. *T. indotineae* is often confused with closely related species *T. mentagrophytes* and *T. interdigitale*, thus **rapid and correct identification** of clinical isolates is vital to achieve successful treatment, as *T. indotineae* strains are often resistant to terbinafine (TBF). Understanding the **resistance profile** of *T. indotineae* strains is very important when choosing a treatment strategy.

RESEARCH



RESULTS

1 MALDI-TOF MS IDENTIFICATION



2 SCREENING MEDIUM

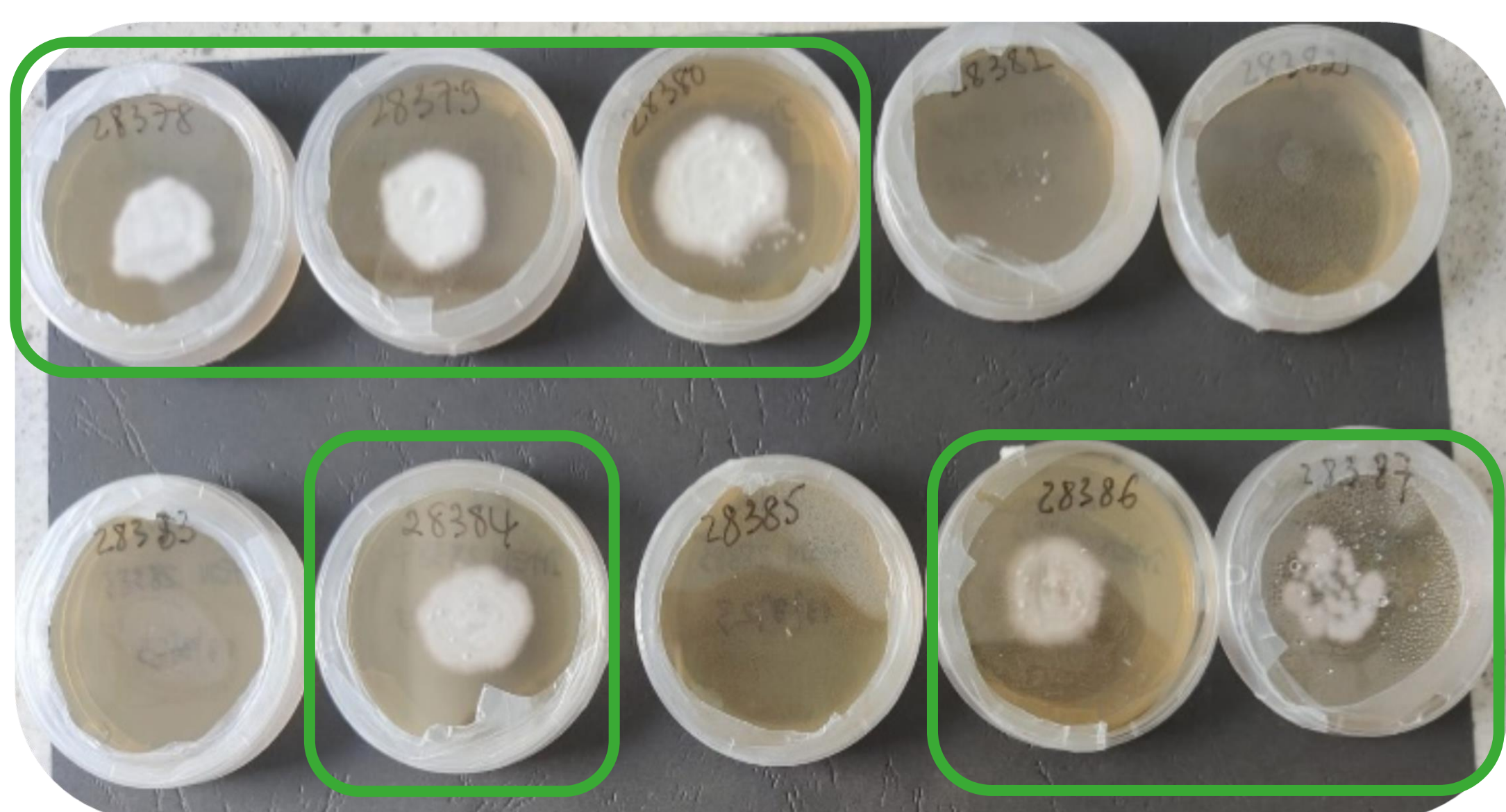


Figure 2. After just 96 hours of incubation (25°C), a clear difference in fungal growth between TBF resistant (green) and sensitive *T. indotineae* strains can be observed on a TBF containing screening medium.

3 ANTIFUNGAL SUSCEPTIBILITY TESTING

Antifungal agent	Geometric mean of the minimal inhibitory concentration (MIC, mg/L)		
	All <i>T. indotineae</i> strains (n = 20)	TBF susceptible strains (n = 11)	TBF resistant strains (n = 9)
Fluconazole	33,13	30,05	37,33
Itraconazole	0,16	0,17	0,15
Voriconazole	0,25	0,25	0,25
Ketoconazole	0,31	0,32	0,29
Terbinafine	0,67	0,05*	> 16,00*
Griseofulvin	3,48	2,92	4,32
Ciclopirox olamine	0,71	0,73	0,68
Naftifine	1,75	0,17*	29,63*
Amorolfine	0,12	0,09*	0,17*

Table 1. Geometric means of MIC-values, with statistically significant (*) ($p = 0,05$) augmentations in TBF resistant *T. indotineae* strains compared to susceptible strains for antifungal agents naftifine and amorolfine.

4 INVESTIGATING AMINO ACID SUBSTITUTIONS IN SQLE

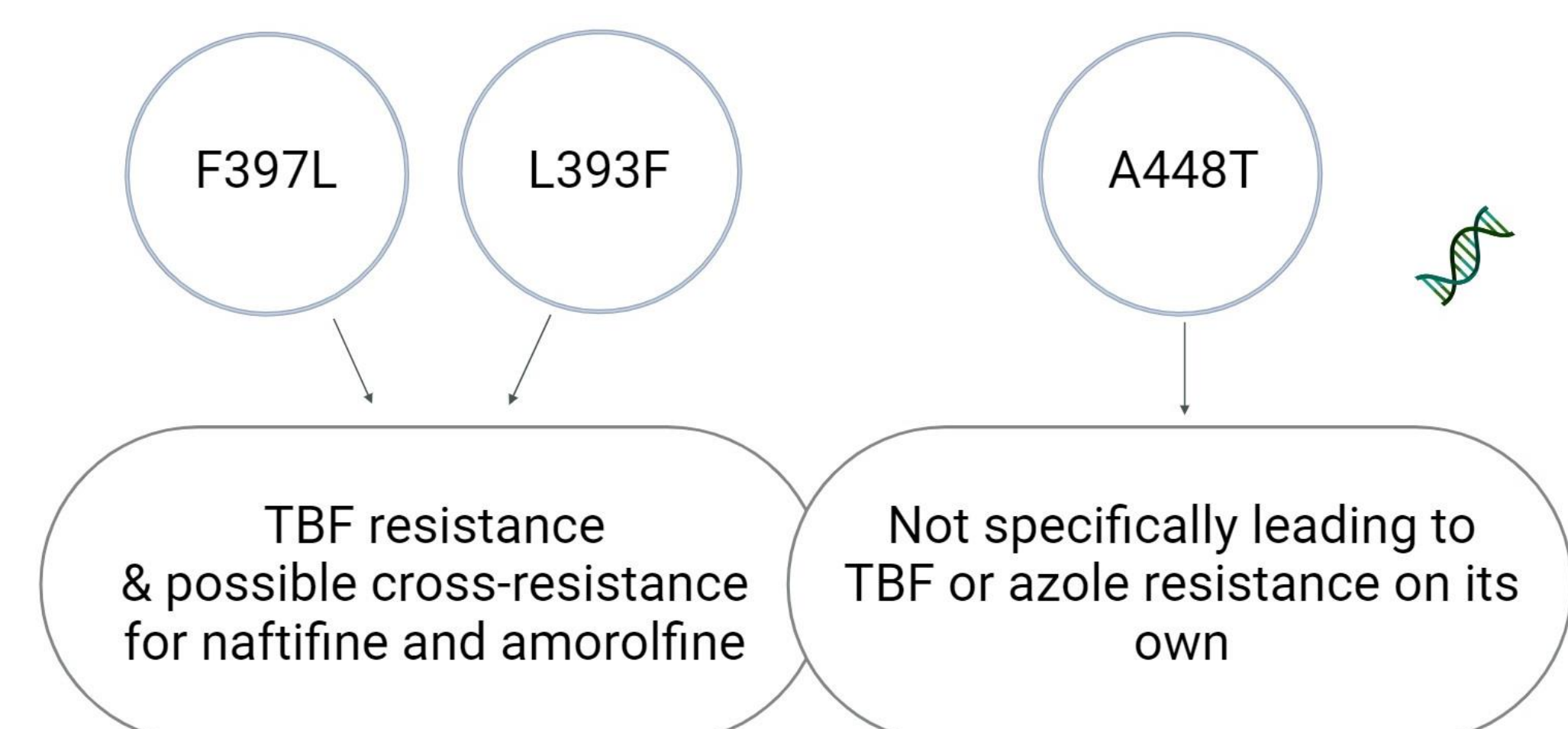


Figure 3. Found amino acid substitutions in *SQLE* and their consequences.

CONCLUSIONS

1. MALDI-TOF MS is a strong and reliable identification tool for *T. indotineae* strains, as it is able to differentiate this pathogen from closely related species.
2. Nevertheless, MALDI-TOF MS could not make a distinction between TBF susceptible and resistant *T. indotineae* strains.
3. Suspicion of cross-resistance between TBF and antifungal agents naftifine and amorolfine, as significantly elevated MIC-values were observed for these drugs.
4. TBF resistance was caused by point mutations in *SQLE*, mainly resulting in amino acid substitutions F397L and L393F, while amino acid substitution A448T does not seem to cause (azole) resistance in *T. indotineae* when no other substitutions are present.